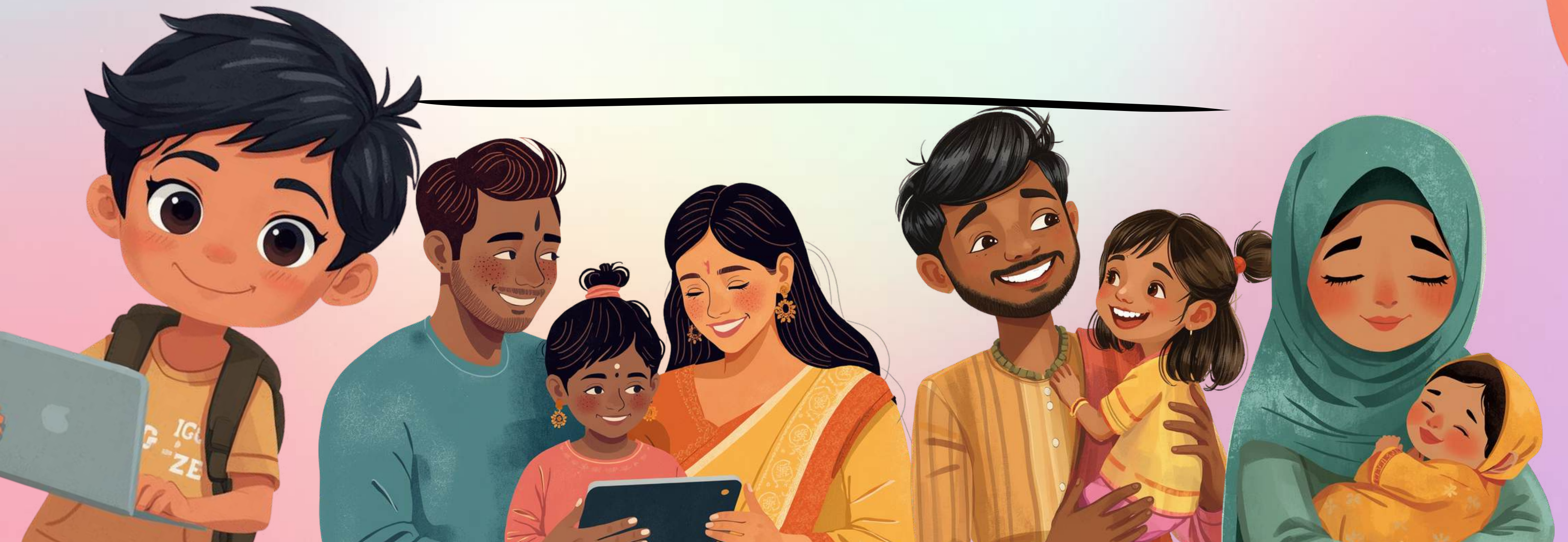


PYTHON'S MAGIC TOOLBOXES: MODULES & THE STANDARD LIBRARY



IMAGINE A MASALA DABBA! 🥘

A Masala Dabba is a kitchen toolbox 🥘

It has all the spices you need to cook tasty food!

You don't grow spices every time, you just take what you need!



WHAT IS A PYTHON MODULE?

A module is a special toolbox in Python



It has ready-made tools (code) to help us!

Like a Masala Dabba, we 'import' only what we need.

Aarav uses the math module to help with numbers



USING THE 'IMPORT' KEYWORD



To use a module, we write: `import module_name` 🔑

Example: `import math`

This brings the math toolbox to your program!

Code:

```
import math
print(math.sqrt(16)) # Diya finds the square
root of 16
```


MEET THE MATH MODULE

math helps with special
calculations ÷

Example: Calculate area of a
Rangoli circle 🎨

We use `math.pi` and `math.sqrt`
Arjun calculates Rangoli area
using `math.pi`



MEET THE RANDOM MODULE

The random module helps us make choices 🎲

Example: Who bats first in gully cricket? 🏏

We use `random.choice()` for a digital coin toss!

Aarav uses random to pick who bats first



HANDS-ON ACTIVITY: GULLY CRICKET TOSS



Let's code a digital coin toss! Type this program:

```
import random
```

```
players = ['Aarav', 'Diya', 'Arjun']
```

```
print(random.choice(players), "bats first!")
```

✨ Try it! Add your friends' names to the list and run again!

MATH MODULE ACTIVITY: RANGOLI GARDEN SIDE LENGTH

Let's find the side length of a square Rangoli garden! 🎨 We know the area is 81 square meters. Use `math.sqrt` to find the side!

```
import math
```

```
area = 81 # in square meters
```

```
side = math.sqrt(area)
```

```
print("Side length is", side, "meters") # Diya measures garden
```


WHAT IS THE STANDARD LIBRARY?

- Python comes with many toolboxes called the Standard Library 📖
- Like a First Aid Kit in every Indian school bus 🚌
- It has tools ready to help you anytime!
- No need to build everything from scratch 🎉



QUICK QUIZ! WHAT DID WE LEARN?

1

What is a
Python module?



2

How do we
bring a module
into our
program? 

3

Name one thing
math or random
module can do!  ÷



SUMMARY & KEEP EXPLORING! ✨

Modules are magic
toolboxes in Python!

Use 'import' to bring in
your helpers

math and random
modules make coding
fun and easy! 🎲 ÷

Keep practicing with
Aarav, Diya & Arjun!



THANK YOU! SEE YOU IN SESSION 2 ✨

Get ready for more Python magic next time!
PLANRS Academy

